

Memory Integrated Tool with Confidence Middleware Layer

Decision layer that a scaffolded AI agent would actually use at runtime to decide:

"Do I clarify, comply, reframe, or produce something they didn't ask for but actually need?"

And the governing prior is: the user is probably wrong about what they want, but always right about what they feel.

Confidence Middleware

A module that sits between user input and the LLM. It scores the input, updates priors from session history, then routes: clarify / comply / reframe / produce the better thing. Returns a decision object your scaffold consumes.

Memory Integrated Tool

The Bayesian engine is part of memory. It persists per-user, per-session, tracking bias patterns over time. Each interaction writes back to the memory scaffold. The agent's "personality" shifts to more assertive, less compliant as it accumulates evidence the user doesn't know what they want.

User input

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[Bayesian Middleware] ← reads memory store (JSON/SQLite)

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decision: {route, confidence, bias_detected, reframe}

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[Prompt Builder] – injects decision into system prompt

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Ollama LLM

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Response + posterior write-back → memory store

memory_store.py

The persistent brain. Saves human_prior, ai_prior, agent_assertiveness, bias fingerprint, and a rolling 10-entry window to scaffold_memory.json. The assertiveness score drifts upward every time the AI gets a wrong or clarify outcome; it literally gets bolder the more it gets burned by vague human input.

bayesian_middleware.py

The decision layer. Call pre_process(user_message) before Ollama, get back a Decision object with route and a ready-made system_injection string. Call post_process(response, decision) after it infers the outcome and writes posteriors back. Four routes: comply, reframe, clarify, challenge. The system prompt injected into Ollama changes depending on which route fires.

scaffold_example.py

Minimal working loop. Drop your model name in, run it, type snapshot at any prompt to dump the current memory state.

monitor.html

Drag your scaffold_memory.json onto it and it renders your live session: belief posteriors, bias radar, outcome distribution, assertiveness dial. Also works standalone as a manual simulator.

Note: the bias classifier is keyword heuristics right now. It's fast and zero-dependency, but crude.

If tiered memory system already has an embedding layer running through Ollama, potential to swap it for a proper semantic classifier trivially: `classify_bias()` in the middleware is a clean seam.